

ER Physiology Practice Questions with answers

Topics: Hypothalamus, Anterior Pituitary & Posterior Pituitary

1. A 39-year-old woman comes to the physician because of an 11-month history of enlargement of her hands and feet, 6-kg (13.2-lb) weight gain, and difficulty seeing things to the right and left of her. Laboratory studies show elevated serum growth hormone levels. This hormone induces production of a second hormone to achieve many of its growth promoting effects. Which of the following organs most likely secretes this second hormone?

- A. Stomach
- B. Kidney
- C. Hypothalamus
- D. Parathyroid gland
- E. Liver

2. A 26-year-old woman comes to the physician because of missing her period. She is tested positive for pregnancy. The hormone detected in her blood and urine was found to be human chorionic gonadotropin (hCG). Which of the following hormones of the anterior pituitary most likely belongs to the same family as hCG?

- A. Adrenocorticotrophic hormone
- B. Luteinizing hormone
- C. Growth hormone
- D. Prolactin
- E. Oxytocin

3. A 41-year-old man comes to the physician because of a 9-month history of enlargement of his fingers and toes, increase in size of his shoes, and occasional headaches. He has no history of major medical illness. His blood pressure is 146/92 mm Hg; other vital signs are within normal limits. Physical examination shows enlarged and protruding lower jaw, prominent maxilla, frontal bossing, and enlarged fingers and toes. Which of the following laboratory findings is most likely in this patient?

- A. Elevated serum glucose
- B. Low serum calcium
- C. Decreased serum free fatty acids
- D. Increased urinary amino acids
- E. Elevated serum prolactin

4. A 34-year-old woman comes to the physician because of a 4-month history of missing her menstrual periods. She was diagnosed with some mental illness six months ago and takes an anti-psychotic drug. Physical examination shows mild breast tenderness and nipple discharge. Pregnancy test is negative. Increased blood level of which of the following is the most likely cause of this patient's presentation?

- A. Dopamine
- B. Gonadotropin releasing hormone
- C. Oxytocin
- D. Prolactin
- E. Thyroid-stimulating hormone

5. A 52-year-old man comes to the physician because of an 8-month history of headaches, feeling cold and difficulty concentrating in his work. His temperature is 36.3°C (97.4°F), pulse is 48/min, and blood pressure is 128/86 mm Hg. Laboratory studies show low serum levels of T3 and T4. MRI of the brain shows a mass in the sella turcica. A deficiency of which of the following hormones is most likely responsible for this patient's presentation?

- A. Anti-diuretic hormone
- B. Gonadotropins
- C. Thyrotropin-releasing hormone
- D. Thyroid-stimulating hormone
- E. Cortisol

6. An investigator is studying the effect of a new drug. In the experiment, rats are injected with this drug, which causes secretion of a protein hormone by the liver. The investigator observes increased chondrogenesis at epiphyseal plates and suppression of the growth hormone secretion. Which of the following additional effects is most likely by the protein secreted by the liver?

- A. Decreases blood glucose
- B. Increases amino acid uptake by the muscles
- C. Increases breakdown of fat
- D. Increases glucose uptake by the muscles
- E. Decreases lean body mass

7. A 56-year-old man comes to the physician because of a 4-month history of frequent urination and increased thirst. Physical examination shows dehydration. Laboratory studies show:

Serum glucose (fasting): 96 mg/dL (Normal: 70-100),
Serum osmolality: 314 mOsm/kg (Normal: 275-295),

Urine osmolality: 150 mOsm/kg (Normal: 50-1200).

Urine osmolality measured after 4 hours of water deprivation is 152 mOsm/kg. He is administered with a synthetic vasopressin injection and the urine osmolality measured 30 minutes after this injection is 650 mOsm/kg. Which of the following structures is most likely abnormal in this patient?

- A. Anterior pituitary
- B. Adrenal cortex
- C. Adrenal medulla
- D. Supraoptic nucleus of the hypothalamus
- E. Ventromedial nucleus of the hypothalamus

8. A 61-year-old woman comes to the physician because of a 3-week history of increased thirst and frequent urination with large volume of urine. He had brain surgery to remove a pituitary tumor 4 weeks ago. Laboratory studies show undetectable serum vasopressin level and a urine osmolality of 126 mOsm/kg. Which of the following physiological effects on the renal tubules is most likely by the deficient hormone?

- A. Decreases water permeability
- B. Inhibition of phospholipase C formation
- C. Increases chloride channels
- D. Increases in cAMP production
- E. Stimulation of phospholipase A

9. A 26-year-old healthy man volunteers for a research study in which he is administered 1 L of 3% dextrose in water (a hypotonic solution) as an intravenous infusion. The physiological effects of this infusion on kidneys and brain are assessed. Which of the following effects is most likely to occur in this subject after this infusion?

- A. Decreased reabsorption of urea from the inner medullary collecting duct
- B. Increased aquaporin channels in the collecting duct
- C. Increased plasma osmolality
- D. Increased renal glucose excretion
- E. Increased secretion of posterior pituitary hormones
- F. Stimulation of osmoreceptors in the hypothalamus

10. A 26-year-old woman comes to the physician because of a 5-month history of persistent inability to breastfeed after giving birth of her first child. She also has severe fatigue, weight loss of 7-kg (15.4-lb), and cold intolerance. She says that her delivery was complicated by excessive blood loss, and she received 3 units of packed red blood cell infusion. Laboratory studies show a fasting blood glucose of 65 mg/dL (normal: 70-

100) and low levels of ACTH, LH, FSH, TSH and growth hormone. Which of the following additional hormones is most likely deficient in this patient?

- A. Vasopressin
- B. Oxytocin
- C. Prolactin
- D. Dopamine
- E. Glucagon

11. A study is conducted using transgenic mice that cannot secrete neurophysin. These mice are raised to become adults. At the age of 16 weeks, a blood sample is drawn, and levels of various electrolytes are measured. Which of the following serum abnormalities is most likely in these mice?

- A. Hypernatremia
- B. Hyperkalemia
- C. Hypokalemia
- D. Hypocalcemia
- E. Hyponatremia

12. A 24-year-old primigravid woman at 36 weeks' gestation comes to the emergency department because of a 6-hour history of labor pain. Her vital signs are within normal limits. Physical examination shows normal fetal heart sounds. Vaginal examination shows softening of the cervix which is 6-cm dilated. The stretch of the cervix increases the hormone in her blood by a positive feedback mechanism. Cells in which of the following structures most likely synthesize this hormone?

- A. Uterine smooth muscles
- B. Anterior pituitary gland
- C. Hypothalamus
- D. Mammary glands
- E. Posterior pituitary gland

13. A 58-year-old woman comes to the physician because of a 2-week history of headache, increased volume, and frequency of urination. She says that she drinks a large volume of fluids because of increased thirst. Her vital signs are within normal limits. Physical examination shows no abnormalities. Laboratory studies show:
Serum sodium concentration: 149 mEq/L (N: 135-145),
Serum osmolality: 305 mOsm/kg (N: 280-290),
Urine osmolality: 150 mOsm/kg,

Urine glucose: Absent.

MRI shows a mass in the hypothalamic region. Which of the following additional laboratory findings is most likely in this patient?

- A. A negative free water clearance
- B. Decreased serum aldosterone
- C. Elevated serum vasopressin
- D. A positive free water clearance
- E. Osmotic diuresis

14. A 41-year-old man comes to the physician because of a 5-week history of fatigue, dizziness, and cold intolerance. He was diagnosed with a pituitary tumor and underwent surgery one month ago. His pulse is 92/min and blood pressure is 96/66 mm Hg.

Physical examination shows dehydration. Laboratory studies show a fasting glucose level of 65 mg/dL and decreased serum thyroid hormones and cortisol. Which of the following hormone levels is most likely decreased in this patient?

- A. Adrenocorticotrophic hormone
- B. Corticotropin-releasing hormone
- C. Insulin
- D. Parathyroid hormone
- E. Thyrotropin-releasing hormone

15. A 31-year-old woman comes to the physician because of an 8-month history of headaches, menstrual irregularity, low sexual drive, and defective vision. She says she wishes to have a child and is not taking any contraceptive pills. Her vital signs are within normal limits. Physical examination shows a milky discharge from her breasts.

Laboratory studies show extremely low levels of gonadotropins (LH and FSH). MRI of brain shows pituitary mass. Which of the following hormones is most likely elevated in this patient?

- A. Oxytocin
- B. Progesterone
- C. Estrogen
- D. Androgens
- E. Prolactin

ANSWER KEY AND EXPLANATION

1. A 39-year-old woman comes to the physician because of an 11-month history of enlargement of her hands and feet, 6-kg (13.2-lb) weight gain, and difficulty seeing things to the right and left of her. Laboratory studies show elevated serum growth hormone levels. This hormone induces production of a second hormone to achieve many of its growth promoting effects. Which of the following organs most likely secretes this second hormone?

- A. Stomach
- B. Kidney
- C. Hypothalamus
- D. Parathyroid gland
- E. Liver

Answer: E. This patient has acromegaly (GH excess). GH stimulates production of insulin like growth factor-1 (IGF-1) in the liver. IGF-1 mediates growth promoting effects of the GH.

2. A 26-year-old woman comes to the physician because of missing her period. She is tested positive for pregnancy. The hormone detected in her blood and urine was found to be human chorionic gonadotropin (hCG). Which of the following hormones of the anterior pituitary most likely belongs to the same family as hCG?

- A. Adrenocorticotropic hormone
- B. Luteinizing hormone
- C. Growth hormone
- D. Prolactin
- E. Oxytocin

Answer: B. hCG is a hormone detected during the early period of pregnancy and it is a marker used in pregnancy tests. This hormone belongs to the glycoprotein family of hormones. This hormone is structurally similar to the glycoprotein hormones secreted by the anterior pituitary, namely, follicle-stimulating hormone (FSH), luteinizing hormone (LH), and thyroid stimulating hormone (TSH). All have the same α subunit; it is their β subunit that confers functional specificity.

3. A 41-year-old man comes to the physician because of a 9-month history of enlargement of his fingers and toes, increase in size of his shoes, and occasional headaches. He has no history of major medical illness. His blood pressure is 146/92

mm Hg; other vital signs are within normal limits. Physical examination shows enlarged and protruding lower jaw, prominent maxilla, frontal bossing, and enlarged fingers and toes. Which of the following laboratory findings is most likely in this patient?

- A. Elevated serum glucose
- B. Low serum calcium
- C. Decreased serum free fatty acids
- D. Increased urinary amino acids
- E. Elevated serum prolactin

Answer: A. This patient has acromegaly, a clinical condition of elevated growth hormone (GH) in adults. The clinical features of acromegaly include large and protruding mandible (macrognathia), large tongue (macroglossia), enlargement of the hands and feet, and hypertension. GH reduces glucose uptake by the peripheral tissues and thus elevates blood glucose and may result in diabetes mellitus.

4. A 34-year-old woman comes to the physician because of a 4-month history of missing her menstrual periods. She was diagnosed with some mental illness six months ago and takes an anti-psychotic drug. Physical examination shows mild breast tenderness and nipple discharge. Pregnancy test is negative. Increased blood level of which of the following is the most likely cause of this patient's presentation?

- A. Dopamine
- B. Gonadotropin releasing hormone
- C. Oxytocin
- D. Prolactin
- E. Thyroid-stimulating hormone

Answer: D. Common symptoms of prolactin excess are amenorrhea, decreased sexual drive, and milk secretion from breast (galactorrhea). Antipsychotic drugs inhibit dopamine secretion and thus disinhibit the negative effect of dopamine on prolactin secretion. This results in elevated serum prolactin.

5. A 52-year-old man comes to the physician because of an 8-month history of headaches, feeling cold and difficulty concentrating in his work. His temperature is 36.3°C (97.4°F), pulse is 48/min, and blood pressure is 128/86 mm Hg. Laboratory studies show low serum levels of T3 and T4. MRI of the brain shows a mass in the sella turcica. A deficiency of which of the following hormones is most likely responsible for this patient's presentation?

- A. Anti-diuretic hormone
- B. Gonadotropins
- C. Thyrotropin-releasing hormone
- D. Thyroid-stimulating hormone
- E. Cortisol

Answer: D. This patient has a tumor in the pituitary gland. A tumor may lead to decreased or increased secretion of one or more hormones of the anterior pituitary. This patient has clinical and laboratory features of reduced thyroid hormones and therefore, the tumor might have reduced secretion of TSH from the anterior pituitary.

6. An investigator is studying the effect of a new drug. In the experiment, rats are injected with this drug, which causes secretion of a protein hormone by the liver. The investigator observes increased chondrogenesis at epiphyseal plates and suppression of the growth hormone secretion. Which of the following additional effects is most likely by the protein secreted by the liver?

- A. Decreases blood glucose
- B. Increases amino acid uptake by the muscles
- C. Increases breakdown of fat
- D. Increases glucose uptake by the muscles
- E. Decreases lean body mass

Answer: B. The protein secreted by the liver by the new drug is IGF-1. Direct actions of IGF-1 include growth promotion, increasing amino acid uptake, favoring protein synthesis and positive nitrogen balance.

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- A. Anterior pituitary
- B. Adrenal cortex
- C. Adrenal medulla
- D. Supraoptic nucleus of the hypothalamus
- E. Ventromedial nucleus of the hypothalamus

Answer: D. This patient has a deficiency of vasopressin secretion from the hypothalamus through posterior pituitary. This condition is called diabetes insipidus. The type of diabetes insipidus in this patient is called neurogenic or central type as he has deficiency of vasopressin synthesis in the brain.

8. A 61-year-old woman comes to the physician because of a 3-week history of increased thirst and frequent urination with large volume of urine. He had brain surgery

to remove a pituitary tumor 4 weeks ago. Laboratory studies show undetectable serum vasopressin level and a urine osmolality of 126 mOsm/kg. Which of the following physiological effects on the renal tubules is most likely by the deficient hormone?

- A. Decreases water permeability
- B. Inhibition of phospholipase C formation
- C. Increases chloride channels
- D. Increases in cAMP production
- E. Stimulation of phospholipase A

Answer: D. This patient has diabetes insipidus because of a deficiency of vasopressin. Vasopressin acts on the principal cells of the collecting duct in the kidney. Here, vasopressin binds to the V2 receptors and stimulates adenylyl cyclase to produce cAMP as the second messenger.

9. A 26-year-old healthy man volunteers for a research study in which he is administered 1 L of 3% dextrose in water (a hypotonic solution) as an intravenous infusion. The physiological effects of this infusion on kidneys and brain are assessed. Which of the following effects is most likely to occur in this subject after this infusion?

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Answer: A. Infusion of a hypotonic solution decreases osmolality of plasma. This decrease in serum osmolality inhibits osmoreceptors and thus inhibits vasopressin secretion. Low serum vasopressin causes diuresis and decreases urea reabsorption in the collecting duct. Also, reabsorption of urea in the inner medullary collecting duct is dependent on vasopressin and in the absence of vasopressin, this effect adds up.

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- A. Vasopressin
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Answer: C. This patient has panhypopituitarism in which all anterior pituitary hormones are deficient. The blood loss during labor had caused ischemic necrosis of the pituitary gland. This condition is called Sheehan syndrome (postpartum ischemic necrosis of the pituitary gland). Posterior pituitary hormones may be normal as the blood supply to this part is by a different blood vessel.

11. A study is conducted using transgenic mice that cannot secrete neurophysin. These mice are raised to become adults. At the age of 16 weeks, a blood sample is drawn, and levels of various electrolytes are measured. Which of the following serum abnormalities is most likely in these mice?

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Answer: A. These mice cannot synthesize posterior pituitary hormones, vasopressin, and oxytocin because of deficiency of neurophysin. Deficiency of vasopressin produces neurogenic diabetes insipidus and loss of water in urine. This results in hypernatremia.

12. A 24-year-old primigravid woman at 36 weeks' gestation comes to the emergency department because of a 6-hour history of labor pain. Her vital signs are within normal limits. Physical examination shows normal fetal heart sounds. Vaginal examination shows softening of the cervix which is 6-cm dilated. The stretch of the cervix increases the hormone in her blood by a positive feedback mechanism. Cells in which of the following structures most likely synthesize this hormone?

- A. Uterine smooth muscles
- B. Anterior pituitary gland
- C. Hypothalamus
- D. Mammary glands
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Answer: C. The hormone secreted by cervical dilation is oxytocin. Oxytocin is synthesized from supraoptic and paraventricular neurons of the hypothalamus and stored in posterior pituitary.

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- A. A negative free water clearance
- B. Decreased serum aldosterone
- C. Elevated serum vasopressin
- D. A positive free water clearance
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Answer: D. This patient has neurogenic (central) diabetes insipidus because of a brain tumor. In patients with central DI, the kidneys make large amounts of diluted urine due to a lack of vasopressin. All patients with diabetes insipidus are associated with low urine osmolality and positive free water clearance. If the patient does not drink enough fluid (to compensate for the urinary loss), the plasma volume decreases which leads to hypernatremia. As this patient drinks isotonic fluid, her vital signs are normal, and her serum sodium is within normal range.

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- E. Thyrotropin-releasing hormone

Answer: A. This patient has panhypopituitarism because of pituitary surgery. Hypopituitarism leads to decreases in pituitary hormones such as adrenocorticotrophic hormone, thyroid-stimulating hormone, follicle-stimulating hormone, luteinizing hormone, GH, and prolactin. Hypothalamic hormones are not decreased.

15. A 31-year-old woman comes to the physician because of an 8-month history of headaches, menstrual irregularity, low sexual drive, and defective vision. She says she wishes to have a child and is not taking any contraceptive pills. Her vital signs are within normal limits. Physical examination shows a milky discharge from her breasts.

Laboratory studies show extremely low levels of gonadotropins (LH and FSH). MRI of brain shows pituitary mass. Which of the following hormones is most likely elevated in this patient?

- A. Oxytocin
- B. Progesterone
- C. Estrogen
- D. Androgens
- E. Prolactin

Answer: E. This patient has pituitary prolactinoma. Most common clinical features of high serum prolactin include menstrual irregularities, loss of interest in sex, infertility, and galactorrhea. Elevated serum prolactin inhibits the secretion of gonadotropin releasing hormone of the hypothalamus, resulting in low serum levels of LH and FSH.